

Frank — AMC Upper Primary **STRETCH TAIL Q26–Q30**

30 Jun 2026 | The 5 integer-answer tail questions that decide a Prize. These are pitched at (or **above**) the real-paper ceiling — trial-and-error will not finish them in time. Whole-number answers; show your working.

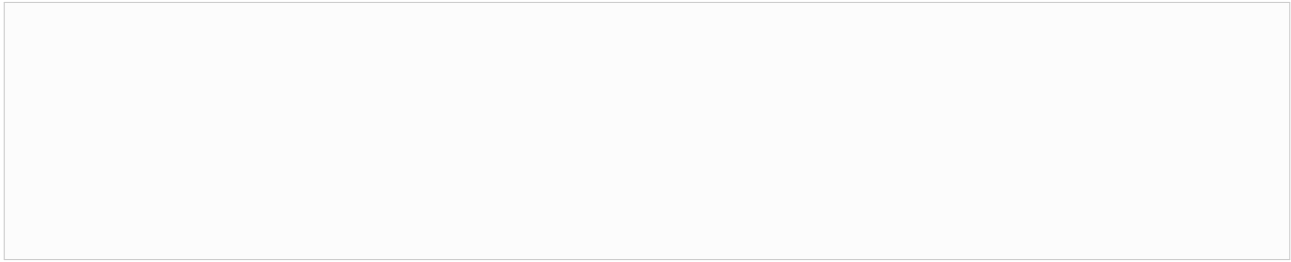
Q26. [L2] [7 marks] What is the smallest three-digit number with all of these properties: divisible by 6; contains the digit 3; has no digit 0; has all different digits; has exactly 1 odd digit?

Q27. [L3] [7 marks] In how many ways can you choose 3 different numbers from 2 to 22 so that their sum is a multiple of 3?

Q28. [L3] [8 marks] How many four-digit numbers have all of these properties: divisible by 12; has all different digits; has no digit 0; has exactly 2 odd digits; has a digit sum divisible by 9; contains the digit 8?

Q29. [L3] [8 marks] You write the 250 numbers 4, 8, 12, ... up to 1000. How many times do you write the digit '0'?

Q30. [L3] [8 marks] How many four-digit numbers have all of these properties: divisible by 13; has all different digits; has no digit 0; has exactly 2 odd digits; has a digit sum divisible by 9; contains the digit 6?



L2 = real-paper difficulty with a hidden condition (don't stop early). L3 = above the real-paper ceiling: the search space is too big to list, so count by structure. Every item is a code-verified twin of a real AMC anchor family (unique answer, difficulty checked in band).